CBOT CORN FUTURES (ZC)

Spread & Curve Structure — Fundamental Reference

US Crop Calendar, Ethanol, China, South America, and the Forward Curve

This document presents a fact-based reference on the fundamental drivers of CBOT Corn futures (ticker: ZC), contract size 5,000 bushels, with particular focus on the forces that shape the forward curve and the spread structure between contract months. Corn is the largest crop grown in the United States by both acreage and value, and the US remains the world's largest producer and (in most years) largest exporter, making the US crop calendar, weather, and the weekly and monthly USDA data cycle the dominant fundamental framework for the CBOT contract. Corn's demand structure is unusually bifurcated relative to other major grains: roughly equal-order-of-magnitude shares of US corn use go to livestock and other animal feed, to ethanol production (a demand category created almost entirely by federal biofuel policy), and to direct food, seed, and industrial uses plus exports — meaning corn fundamentals sit at the intersection of traditional row-crop agriculture, energy policy and the crude oil market via ethanol, and global livestock and protein demand. Understanding the US planting-to-harvest calendar, the weekly and monthly USDA data infrastructure, the ethanol demand pillar, and the growing role of Brazilian safrinha corn as a second global supply season is the foundation of spread and butterfly trading on the corn curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Delivery Mechanics

CBOT Corn is one of the most liquid agricultural futures contracts in the world and the benchmark price reference for global corn and, more broadly, for the feed-grain complex.

Parameter	Detail
Exchange	CME Group — Chicago Board of Trade (CBOT)
Ticker	ZC (front month); historically traded as "C" prior to electronic-ticker convention
Contract size	5,000 bushels (approximately 127 metric tonnes)
Quotation	US cents per bushel (¢/bu); minimum tick = 1/4 cent/bu = $12.50 per contract
Contract months	March (H), May (K), July (N), September (U), December (Z) — five delivery months per year, aligned with the US crop marketing calendar. December is the primary new-crop (harvest) contract; July is the last old-crop contract before new-crop supply becomes deliverable.
Settlement type	Physical delivery via shipping certificate at CBOT-approved grain elevators and shipping stations, primarily along the Illinois River and in the broader Chicago/Midwest river and rail terminal network.
Delivery grade	No. 2 Yellow corn at par; No. 1 Yellow deliverable at a premium, No. 3 Yellow at a discount, per USDA grading standards (test weight, damaged kernels, foreign material, moisture).
Trading hours	CME Globex: Sunday–Friday 19:00–07:45 ET and 08:30–13:20 ET (electronic trading runs nearly continuously with a short daily maintenance break); the 08:30 ET reopening aligns with the US day session and most USDA report releases.
Last trading day	Business day prior to the 15th calendar day of the delivery month
Contract value at $4.50/bu	$22,500 per contract
US marketing year	September 1 – August 31, aligned with the US harvest; the USDA, and most US trade statistics, report on this basis.

1.1 The Old-Crop / New-Crop Structure

As with cotton and other annually-harvested row crops, the distinction between old-crop and new-crop contracts is a central organising feature of the corn curve:

- **Old-crop contracts (March, May, July)**: price corn from the prior autumn's harvest that is currently held in storage (on-farm bins, country elevators, river/export terminals) and being marketed through the remainder of the September–August marketing year. Old-crop supply is fixed at the harvested quantity (subject only to revision as USDA refines its production estimate); old-crop price action is therefore driven primarily by the pace of domestic use and export shipments against that fixed supply.
- **New-crop contracts (September, December of the current year, and March of the following year)**: price corn from the harvest that has not yet occurred or is in progress. New-crop supply is uncertain and depends on planted acreage, growing-season weather, and yield. December is the primary new-crop reference contract, since the bulk of the US harvest reaches elevators and becomes deliverable by early-to-mid December.
- **The July-December spread (old-crop/new-crop)**: directly analogous in function to cotton's July-December spread described elsewhere in this document series. A widening July premium over December (or steep backwardation) signals old-crop tightness; a December premium (contango)

generally reflects comfortable old-crop carryout or anticipated new-crop abundance, though — as with cotton — a wide carry can also reflect the market pricing in a smaller new crop and paying up for deferred delivery, requiring context to distinguish.

2. The US Crop Calendar

The United States plants and harvests corn on a single annual cycle (no multiple cropping within the US growing season, in contrast to some other corn-producing countries and to the US soybean-corn-wheat rotation system more broadly). This single, tightly clustered annual cycle concentrates nearly all of the weather-risk-driven price discovery for the US crop into a relatively narrow window each year.

Period	Stage	Key Weather Risk	Primary Data Release	ZC Curve Impact
Mar–Apr	Planting intentions and early fieldwork preparation	Wet, cold soils delaying fieldwork; spring flooding in river-bottom acreage	USDA Prospective Plantings (late March)	First official acreage signal for new-crop December; a significant single-day market mover
Late Apr–May	Planting	Persistent rain preventing planting progress; cold soil temperatures delaying emergence	USDA weekly Crop Progress (% planted vs 5-yr average)	Slow planting progress raises the risk of delayed maturity and frost exposure at the back end of the season; can shift acreage toward soybeans if planting is delayed past the agronomic corn-planting window
Jun	Emergence and early vegetative growth (V-stages)	Excess rainfall/flooding (saturated soils, nitrogen loss); early-season drought stress	USDA weekly Crop Progress (emergence, condition ratings); USDA June Acreage Report (a major data event, see Section 2.1)	Early condition ratings establish a yield-potential baseline; June Acreage often the most market-moving single event before pollination
Jul	Pollination (silking and tasseling) — the single most yield-critical period	Heat stress (especially overnight low temperatures) and drought during pollination; this is the most weather-sensitive single window in the entire US crop calendar	USDA weekly Crop Progress (silking %, condition ratings); NOAA/CPC short-range temperature and precipitation outlooks	The most volatile weather-trading period of the year for ZC; condition-rating deterioration during pollination has historically produced the sharpest weather-driven price moves
Aug	Grain fill (kernel development)	Continued heat/drought stress reduces kernel weight even after a successful pollination; early frost risk begins to enter market consideration for late-planted crop	USDA August WASDE (first survey-based yield estimate of the season, see Section 2.2)	The August WASDE is typically the single most anticipated scheduled report of the crop year
Sep–Oct	Maturity and harvest	Early frost (kills the plant before maturity, reducing yield and quality); wet harvest conditions delaying fieldwork and degrading grain quality	USDA weekly Crop Progress (harvest %); monthly WASDE updates	Harvest pace confirms or revises the yield estimate; harvest-pressure selling typically weighs on cash basis and can pressure the front of the curve
Nov–Feb	Post-harvest marketing; on-farm and elevator storage	No growing-season weather risk; logistics risk (river levels affecting barge transport, rail service)	USDA weekly export inspections and sales; monthly WASDE	Export pace vs USDA forecast determines whether the marketing-year ending-stocks estimate is on track

2.1 The USDA June Acreage Report

- Published in late June, this report provides USDA's first survey-based (rather than intentions-based) estimate of actual planted corn (and soybean) acreage for the year, refining the March Prospective Plantings intentions figure with on-the-ground farmer-reported data collected after planting is largely complete. Significant deviations from the March intentions figure or from analyst expectations are historically associated with some of the largest single-day moves in the corn futures calendar, since the report changes the assumed size of the entire new crop before a single yield estimate has even been made.

2.2 Pollination Weather — The Most Critical Window

- Corn yield is disproportionately sensitive to weather stress (particularly heat and moisture deficit) during the roughly two-week pollination (silking) window in July, more so than during any other phase of the growing season, because heat stress during pollination can directly reduce the number of kernels that successfully form, an effect that cannot be recovered later in the season even with subsequent favourable weather. This agronomic fact is the reason the market's weather-risk attention concentrates so heavily on July temperature and precipitation forecasts specifically, more than on any other single calendar window.

3. The USDA Data Infrastructure — The Primary Fundamental Framework

No major commodity market is as thoroughly served by a single, centralised, and methodologically transparent government statistical agency as US row-crop agriculture is by the USDA. Understanding the structure and interaction of the USDA's various corn-related reports is foundational to trading the ZC curve.

3.1 WASDE — The Monthly Balance Sheet

- The World Agricultural Supply and Demand Estimates (WASDE), published monthly (typically around the 11th, though the exact date shifts around the quarterly Acreage/Quarterly Stocks and other major report dates), provides US and world corn production, the demand breakdown (feed and residual use, ethanol/fuel use, food/seed/industrial use, and exports), and ending stocks for both the current and prior marketing years. The US stocks-to-use ratio (ending stocks as a percentage of total annual use) derived from WASDE is the primary structural price-level indicator for corn, analogous to the role of the same ratio for cotton, sugar, and other row crops described elsewhere in this document series.
- **The August, September, and October WASDE reports** carry particular weight because they are the first reports to incorporate USDA's own survey-based yield estimate (rather than a trend-yield assumption used in earlier-year reports), making them the reports most likely to produce a substantial revision to the production estimate.

3.2 NASS Crop Production and Yield Surveys

- USDA's National Agricultural Statistics Service (NASS) conducts the underlying farmer surveys and (from August onward) field-based objective yield measurements (physically counting ears and kernels in sample fields) that feed into the WASDE's yield estimate. The methodology shift from a trend-line yield assumption (used in the spring/early-summer WASDE reports, before any actual yield data exists) to survey- and field-measurement-based estimates (from August onward) is itself an important structural feature of the annual data calendar: early-season WASDE yield figures carry materially more uncertainty than the August-onward figures.

3.3 Weekly Crop Progress and Condition Ratings

- Published every Monday afternoon during the growing season (typically April through November), the weekly Crop Progress report provides planting/emergence/silking/harvest percentage complete (versus the prior year and the 5-year average) and a weekly crop condition rating (Excellent/Good/Fair/Poor/Very Poor) for major producing states. The percentage of the crop rated Good or Excellent (G+E) is the single most widely quoted weekly figure in the agricultural trade press and is tracked as a running, high-frequency proxy for the eventual yield outcome, though the rating is a subjective field-level assessment rather than a yield measurement and can diverge from the eventual survey-based yield figure.

3.4 Quarterly Grain Stocks Reports

- Published quarterly (with the most closely watched editions released at the end of the marketing year in September and at the end of the calendar year in December/January), the Grain Stocks report provides an independent, survey-based measurement of actual corn held in storage (on-farm and off-farm) as of the survey date. Because this figure is measured independently of the WASDE's balance-sheet calculation (production minus cumulative use), a significant divergence between the two — commonly referred to in the trade as a "stocks surprise" — can imply that USDA's demand estimates (feed use, ethanol use, or exports) have been mismeasured and require revision, making the Grain Stocks report a periodically very market-moving release despite its less frequent (quarterly) publication schedule.

4. Ethanol — The Policy-Created Demand Pillar

Corn-based ethanol is the single demand category that most distinguishes corn from other major grains, and exists in its current scale almost entirely as a consequence of US federal biofuel policy rather than as an organically emerged market.

4.1 The Renewable Fuel Standard (RFS)

- The Renewable Fuel Standard, established by the Energy Policy Act of 2005 and substantially expanded by the Energy Independence and Security Act of 2007, sets annual Renewable Volume Obligations (RVOs) mandating minimum volumes of renewable fuel (the large majority of which, in practice, is corn-based ethanol) to be blended into the US transportation fuel supply. As described in greater detail in the RBOB reference document in this series (which covers the RFS and the RIN compliance mechanism from the gasoline-blending side), the RFS is the primary policy lever creating and sustaining corn ethanol demand at its current scale.
- **EPA's annual RVO-setting process**: the EPA proposes and finalises the renewable volume obligation for each compliance year, a process that is periodically the subject of significant industry, agricultural, and refining-sector lobbying and litigation. Changes to the RVO — or to related provisions such as small refinery exemptions (which reduce the effective demand for RINs and, indirectly, for the ethanol that generates them) — are policy events tracked by corn market participants as a direct demand-side variable.

4.2 E10, E15, and the Blend Wall

- **E10 (10% ethanol blend)**: the near-universal US gasoline blend standard, compatible with all gasoline-powered vehicles, and the foundation of the large majority of current US ethanol demand.
- **The "blend wall"**: because the large majority of US gasoline is sold as E10, the practical ceiling on total US ethanol demand (absent wider adoption of higher blends) is approximately 10% of total US gasoline consumption — a structural demand ceiling frequently referenced in industry and policy discussions of the ethanol market's growth potential.
- **E15 and higher blends**: EPA approval of E15 for model year 2001+ vehicles, and periodic regulatory actions addressing seasonal E15 sale restrictions (related to the same RVP/volatility regulations described in the RBOB reference document), represent the primary policy pathway by which US ethanol demand could grow beyond the E10 blend wall; the pace and extent of E15 (and higher-blend) retail infrastructure buildout and consumer adoption is a closely watched longer-run demand variable.

4.3 Ethanol Plant Economics and the Crush Margin

- **The ethanol "crush" or production margin**: ethanol plant profitability is determined by the spread between ethanol sale revenue (plus the value of co-products, primarily distillers dried grains with solubles, or DDGS, a high-protein animal feed co-product) and corn input costs plus natural gas (used for process energy) and other operating costs. When this margin is wide, ethanol plants run at high utilisation, supporting corn demand; when compressed, plants may reduce run rates.
- **DDGS as a co-product feeding back into the feed-grain complex**: because DDGS is itself a significant animal feed ingredient (substituting for a portion of corn and soybean meal in livestock rations), the volume of DDGS generated by ethanol production feeds back into the broader feed-grain demand balance, partially offsetting the corn consumed for ethanol from a net feed-grain-complex perspective.
- **Weekly EIA ethanol production and stocks data**: published by the US Energy Information Administration alongside its petroleum data, providing a weekly, near-real-time signal of US ethanol production volume (and, by implication, the pace of corn grind for ethanol) — a relevant cross-check against the WASDE's monthly ethanol-use estimate.

5. Feed Demand and the Livestock Sector Connection

Feed and residual use is corn's largest single demand category in most years, and connects the corn market directly to the cattle, hog, and poultry sectors' own production cycles and profitability — sectors covered in dedicated reference documents elsewhere in this series.

5.1 The Feed-Grain Complex

- Corn is the primary energy (caloric) component of most US livestock and poultry feed rations, typically combined with soybean meal (the primary protein component) and other ingredients including, as noted in Section 4.3, DDGS. Feed demand for corn is therefore directly tied to the size of the US cattle herd, hog inventory, and poultry flock, and to livestock sector profitability (since poor livestock margins can lead to herd/flock liquidation, reducing feed demand, while strong margins support herd expansion and feed demand growth).
- **The cattle cycle**: as described in the Live Cattle reference document in this series, the US cattle herd moves through multi-year expansion and contraction cycles driven by the multi-year biological lag in cattle breeding decisions. The size of the cattle herd at any point in this cycle is a meaningful, slow-moving structural input to corn feed demand.
- **The hog cycle**: as described in the Lean Hog reference document, hog production responds more quickly to relative feed-cost and hog-price profitability signals than cattle, given the shorter hog production cycle, making hog sector feed demand somewhat more responsive to near-term corn price changes than cattle sector feed demand.

5.2 USDA Hogs and Pigs / Cattle Inventory Reports

- USDA's quarterly Hogs and Pigs report and semi-annual Cattle inventory report (both referenced in greater detail in their respective dedicated reference documents in this series) provide the herd/flock-size data that translates into the feed-demand component of the corn balance sheet, making these reports indirectly relevant to corn even though they are not corn-specific publications.

6. Global Production and Trade

While the US dominates the corn export trade in most years and sets the global price-discovery benchmark via CBOT futures, production is geographically diversified across several major producing regions, each with a distinct crop calendar.

Country/Region	Approx. Share of World Production	Crop Calendar	Key Risk	ZC Relevance
United States	~30–33%	Apr–May planting; Jul pollination; Sep–Oct harvest (single annual crop)	Pollination-window heat/drought (Section 2); early frost; spring planting delays	The price-setting benchmark producer and (in most years) the largest exporter; the USDA data infrastructure (Section 3) is built around the US crop.
China	~22–25%	Apr–May planting (varies by region); Sep–Oct harvest	Drought in northeast producing provinces; domestic policy (reserve stock management, import quota administration)	The largest single importer in most recent years; Chinese import demand and state reserve policy are closely tracked, analogous in concept to Chinese demand dynamics described for copper and cotton elsewhere in this series.
Brazil	~9–11%, rapidly growing	Two distinct crops: the "first crop" (planted Sep–Oct, harvested Feb–Mar, often following soybeans in rotation) and the "safrinha" or second crop (planted Jan–Mar immediately after soybean harvest, harvested Jun–Aug)	Late soybean harvest delaying safrinha planting (a critical timing risk — see Section 6.1); drought during the safrinha's typically drier growing window	The safrinha crop has grown to represent the substantial majority of total Brazilian corn output and a globally significant second annual supply pulse — see Section 6.1.
Argentina	~5–6%	Sep–Nov planting; Mar–Jun harvest	Drought; export tax and export quota policy changes (a recurring Argentine agricultural policy variable)	Significant exporter; Argentine government export policy changes are a periodically market-moving variable, analogous to Argentine soybean/wheat policy dynamics.
Ukraine	~3–4% in typical pre-war years, variable since 2022	Apr–May planting; Sep–Oct harvest	War-related disruption to planting, harvesting, and — most significantly — Black Sea export logistics since February 2022	A globally significant exporter whose export logistics have been materially affected by the war in Ukraine; Black Sea shipping corridor status is a tracked variable.
EU, other producers	Remainder	Varies	Varies	Collectively relevant to the global balance but individually smaller than the above origins.

6.1 Brazilian Safrinha Corn — A Structurally Important Second Season

- **What safrinha means**: Portuguese for "little harvest," the safrinha (second-crop) corn is planted immediately following the soybean harvest on the same land, typically January through March, and harvested June through August — a cropping-system innovation that has allowed Brazilian farmers to harvest two crops from the same field within a single growing season.
- **Growth in relative importance**: the safrinha crop has, over the past two decades, grown from a minor supplementary crop to represent the substantial majority of total Brazilian corn production, a structural shift that has made Brazil a globally significant exporter with a harvest season that falls in the Northern Hemisphere summer — providing a second major global corn supply pulse roughly six months out of phase with the US harvest, broadly analogous in structural market function (though operating through a different agronomic mechanism) to the role Brazilian Centre-South cane plays as a second major annual sugar supply event relative to other producing regions.
- **The critical planting-timing risk**: because safrinha corn is planted only after the soybean harvest is complete, any delay to the soybean harvest (itself dependent on the timing of the preceding soybean planting and growing season) compresses the safrinha's growing window and pushes its later growth stages (including its own critical pollination period) further into the typically drier late-season period, raising drought risk for the safrinha crop specifically. This planting-timing dependency on the prior soybean season is a uniquely Brazilian structural risk factor with no direct equivalent in the US single-crop calendar.

7. Export Demand and Logistics

7.1 Major Export Destinations

- **Mexico**: historically the largest single destination for US corn exports, reflecting Mexico's large livestock and food-processing sectors and its geographic proximity and trade-agreement access (USMCA) to US supply.
- **China**: a major and, in some recent years, the largest single destination for US corn exports, subject to the same tariff and trade-policy sensitivity described for other US agricultural exports to China (cotton, soybeans) elsewhere in this document series; Chinese import demand also responds to domestic Chinese harvest size and state reserve stock management decisions.
- **Japan, South Korea, and other Asian destinations**: established, relatively stable import markets, primarily for feed use.

7.2 US Export Logistics — The River and Gulf System

- **The Mississippi River barge system**: the primary domestic transport route moving corn (and soybeans) from Midwest production areas to export terminals at the US Gulf Coast, where the large majority of US corn exports are loaded onto ocean-going vessels.
- **River level risk**: periods of low water on the Mississippi River system (as occurred notably in autumn 2022, a documented low-water episode) constrain barge loading capacity and increase barge freight rates, widening the basis (the difference between local cash price and the CBOT futures price) at interior elevators and temporarily disrupting the normal harvest-season export logistics flow. Conversely, high water/flooding can also disrupt barge operations, though through a different mechanism (lock closures, navigation hazards) than low-water constraints.
- **USDA weekly Grain Transportation Report**: published by USDA's Agricultural Marketing Service, tracking barge freight rates, river conditions, and rail performance — a relevant basis-level data source during periods when river logistics are a live market concern.

8. The Forward Curve Structure — Zones and Spread Mechanics

The corn forward curve's shape is governed by the same cost-of-carry framework as other storable grains (storage cost, financing, and insurance), modulated by the seasonal supply-and-demand pattern dictated by the single annual US harvest, the old-crop/new-crop divide described in Section 1.1, and — increasingly — the partially offsetting timing of the Brazilian safrinha crop.

Contract	Crop-Year Role	Primary Driver	Secondary Driver	Key Signal
December (Z)	New-crop anchor. The primary contract referencing the US harvest just completed.	US planted acreage; growing-season weather (especially July pollination); August WASDE first survey-based yield	Brazilian safrinha harvest outlook (a partially offsetting global supply signal); export sales pace for the marketing year ahead	USDA Prospective Plantings; June Acreage; weekly Crop Progress through pollination; August/September/October WASDE
March (H)	New/old-crop bridge. Post-harvest marketing underway.	US export sales and shipment pace vs USDA forecast; December Grain Stocks report (confirms or revises the production/usage balance)	Early signals on Brazilian safrinha planting progress and pace	Weekly export sales/inspections; December Grain Stocks; early Brazilian planting weather
May (K)	Old-crop. US new-crop planting underway domestically.	US planting pace and early-season weather; Brazilian safrinha crop condition (now in its critical mid-season window)	March Grain Stocks/Prospective Plantings report (released together in late March)	USDA weekly planting progress; Brazilian CONAB safrinha estimates
July (N)	Last old-crop contract. The US new crop is entering its pollination window concurrently.	US pollination-window weather (the single most volatile input of the year, per Section 2.2); old-crop/new-crop spread dynamics (Section 1.1)	Brazilian safrinha harvest results now confirming or revising that crop's contribution to global supply	Daily/weekly US temperature and precipitation forecasts during silking; weekly Crop Progress condition ratings
September (U)	New-crop, early. US harvest beginning in the earliest-maturing regions.	Final pre-harvest condition assessment; early yield indications from the earliest-harvested areas	September WASDE and Grain Stocks (released together, a historically significant data day)	USDA September WASDE; September Grain Stocks report; early harvest yield reports

8.1 Carry, Basis, and Harvest-Pressure

- **Harvest-pressure**: the well-documented seasonal pattern of cash corn basis (and, at times, the futures curve itself) weakening during the September–November harvest window, as the physical market is flooded with newly harvested grain moving off farms and into the elevator system faster than it can be immediately consumed or exported, before working its way higher again as the crop is gradually absorbed into storage and the marketing pipeline over the following months.
- **On-farm storage capacity**: the scale of on-farm grain storage capacity built out across the US Corn Belt over recent decades has, per industry analysis, moderated the severity of harvest-pressure basis weakness compared to earlier periods with less on-farm storage, since farmers have greater ability to store grain on-site and delay selling rather than being forced to deliver immediately at harvest.

9. Macro and Cross-Market Drivers

9.1 Crude Oil and the Ethanol Linkage

- As described in Section 4, corn ethanol economics connect the corn market to crude oil and gasoline prices via the ethanol crush margin: higher gasoline (and by extension crude oil) prices generally support ethanol's blending economics and plant margins, providing a moderately supportive demand-side linkage for corn, though the relationship operates with the RFS mandate (Section 4.1) providing a policy-driven demand floor that somewhat decouples ethanol demand from pure price-based economics relative to a fully market-driven biofuel demand structure.

9.2 The Corn-Soybean Price Ratio and Acreage Competition

- Corn and soybeans compete for the same US farmland, and the relative profitability of planting one versus the other — commonly tracked via the corn-soybean price ratio (soybean price divided by corn price, both per bushel) at the time farmers make planting decisions in the late winter and early spring — influences the acreage split between the two crops reflected in the March Prospective Plantings report and the June Acreage report. A ratio favouring soybean profitability tends to shift marginal acreage toward soybeans and away from corn, and vice versa.

9.3 US Dollar and Export Competitiveness

- As a globally traded, USD-denominated commodity with substantial export demand, corn exhibits the standard dollar-priced-commodity relationship described elsewhere in this document series: USD strength raises the effective cost of US corn for foreign buyers transacting in other currencies, a headwind for US export competitiveness at the margin (particularly relevant given competition from Brazilian and Argentine supply, whose own currencies' moves against the USD affect their relative export pricing).

9.4 Speculative Positioning — COT Data

- **CFTC COT for CBOT Corn**: weekly managed money net positioning is closely tracked, consistent with the positioning framework described throughout this document series. Corn is among the most heavily traded agricultural futures markets, and extreme net long or net short managed money positioning is generally interpreted as a crowding/reversal-risk indicator.
- **Commercial (elevator, processor, exporter) positioning**: grain elevators, ethanol plants, livestock feeders, and exporters hedge forward physical purchase and sale commitments via corn futures and options; the scale and direction of commercial positioning provides context for interpreting the managed money speculative data.

10. Documented Seasonal Patterns

Corn's seasonal pattern is the most calendar-anchored of any market in this document series, given the combination of a single, tightly clustered US annual growing season and the dense, precisely scheduled USDA report calendar built around it.

Period	Crop Stage	Typical ZC Behaviour	Key Variables	Reliability
Jan–Mar (post-harvest)	Marketing of the prior harvest continues. Acreage-intentions data approaching.	Generally calmer trading driven by export pace and the December/March Grain Stocks data, building toward the late-March Prospective Plantings/Stocks data day.	Weekly export sales; quarterly Grain Stocks; corn-soybean price ratio (Section 9.2) ahead of planting decisions	Moderate
Apr–Jun (planting and early growth)	Planting progress; June Acreage report.	Planting-pace and early-condition-rating-driven volatility; June Acreage is a major single-day event.	Weekly Crop Progress (planted %, condition); June Acreage Report	Moderate, rising into June
Jul (pollination)	Silking/tasseling — the single most yield-critical window.	The most volatile weather-trading period of the year; sharp, rapid moves on forecast shifts and confirmed heat/drought stress.	Daily/weekly US temperature and precipitation forecasts; weekly Crop Progress condition ratings	High — the window itself is calendar-certain even though any specific weather outcome is not
Aug (grain fill)	Kernel development; August WASDE (first survey-based yield).	The August WASDE is typically the single most anticipated scheduled report of the year, often producing a significant repricing regardless of direction.	August WASDE; continued condition ratings; early frost-risk monitoring for late-planted crop	High
Sep–Nov (harvest)	Harvest underway and concluding.	Harvest-pressure basis weakness (Section 8.1); September WASDE/Grain Stocks data day; yield confirmation as harvest progresses.	September WASDE and Grain Stocks (same day); weekly harvest progress; basis levels at country elevators	Moderate–High
Dec (year-end)	Post-harvest marketing; December contract approaching expiry.	December Grain Stocks report; year-end fund positioning and index-roll mechanics adding technical flow.	December Grain Stocks; CFTC COT; index roll calendar	Moderate

10.1 Documented Historical Events

- **2012 US drought**: a severe, widespread US Corn Belt drought during the critical July pollination window produced one of the most significant national average yield shortfalls in modern US corn history, with futures prices reaching record nominal highs as the season's weather deterioration was confirmed through successive weekly condition-rating declines and the subsequent August WASDE yield cut.
- **2019 planting-delay year**: persistent spring rainfall across the Corn Belt produced one of the slowest US corn planting paces on record relative to the 5-year average, raising market concern over delayed crop maturity and increased frost exposure; some acreage shifted to soybeans or was left

unplanted (reported via USDA's prevented-planting provisions), illustrating the planting-delay risk channel described in Section 2.

- **Autumn 2022 Mississippi River low-water episode**: documented low water levels on the Mississippi River system during the harvest and export season constrained barge loading capacity and widened export-terminal and interior basis levels, illustrating the river-logistics risk channel described in Section 7.2.
- **Post-2022 Ukraine war disruption to Black Sea exports**: as referenced in Section 6, the war's effect on Ukrainian planting, harvesting, and — most significantly — Black Sea export shipping logistics has been a recurring source of global corn supply and trade-flow uncertainty since February 2022, affecting the global balance alongside the US-centric fundamentals that dominate the CBOT curve.

11. Key Data Sources and Release Schedule

Report	Publisher	Frequency	Typical Timing	Primary ZC Curve Impact
WASDE (Corn component)	USDA WAOB	Monthly	~11th–12th of month, 12:00 ET	All contracts — the primary monthly balance-sheet revision; August–October editions carry the most weight (first survey-based yield)
USDA Prospective Plantings	USDA NASS	Annual	Late March (with March Grain Stocks)	December — first acreage intentions signal; a major single-day mover
USDA Acreage Report (June)	USDA NASS	Annual	Last business day of June	December — confirms/revises planted acreage; historically one of the largest single-day movers of the year
USDA Quarterly Grain Stocks	USDA NASS	Quarterly	Late Mar, Jun, Sep, Jan	Front 1–2 contracts — independent stocks measurement; can reveal demand-estimate errors ("stocks surprise")
USDA Weekly Crop Progress	USDA NASS	Weekly (Apr–Nov)	Monday ~16:00 ET	Front 1–2 contracts during the growing season — G+E rating through pollination is the most-watched single figure
USDA Weekly Export Sales / Inspections	USDA FAS / AMS	Weekly	Thursday 08:30 ET (sales); Monday (inspections)	C1–C3 — export pace vs WASDE forecast; China-specific sales are particularly watched
NOAA/CPC 6–10 and 8–14 day outlooks	NOAA CPC	Daily/twice-weekly	Continuous	Front contract, acutely during Jul pollination — the most weather-headline-sensitive period of the year
EIA Weekly Ethanol Production/Stocks	US EIA	Weekly	Wednesday 10:30 ET	C1–C4 — corn-grind-for-ethanol cross-check against WASDE's monthly ethanol-use estimate
CONAB Brazilian crop estimates (corn component)	CONAB (Brazil)	Monthly	~3rd week of month	C2–C6 — Brazilian safrinha production estimate; a globally significant second-season supply signal
USDA Grain Transportation Report	USDA AMS	Weekly	Thursday	Basis-level — barge freight rates and river conditions, relevant during logistics-disruption episodes
CFTC Commitments of Traders	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — all contracts; extreme managed money positioning as contrary indicator
USDA Hogs and Pigs / Cattle Inventory reports	USDA NASS	Quarterly (hogs) / semi-annual (cattle)	Per report schedule	C4–C12 — feed-demand input via livestock herd/flock size; covered in detail in the respective livestock reference documents

CBOT Corn is, more than almost any other market in this document series, a calendar-anchored, data-infrastructure-driven market: the dense, precisely scheduled USDA report calendar built around a single, tightly clustered US growing season means that corn's fundamental price discovery is concentrated into a small number of known, recurring dates each year — the March Prospective Plantings, the June Acreage Report, the July pollination-weather window, and the August-through-October WASDE reports — to a degree that has few parallels among the other commodities covered in this series. Layered onto this US-centric calendar is a policy-created demand pillar (ethanol, via the RFS) that connects corn to the energy complex, and a growing second annual global supply pulse from Brazilian safrinha production that has, over the past two decades, become structurally important to the global balance.

For spread traders, the most reliable recurring catalysts are the known USDA report dates themselves (rather than weather forecasts in isolation, which are inherently less calendar-certain), the July pollination-window weather risk as the single most volatile period of the year, the old-crop/new-crop (July-December) spread as a structural read on current-marketing-year tightness, and the interaction between US new-crop prospects and the Brazilian safrinha crop's contribution to the global balance as the primary cross-hemisphere supply consideration.